

# MICROPROCESSOR BASED SYSTEMS

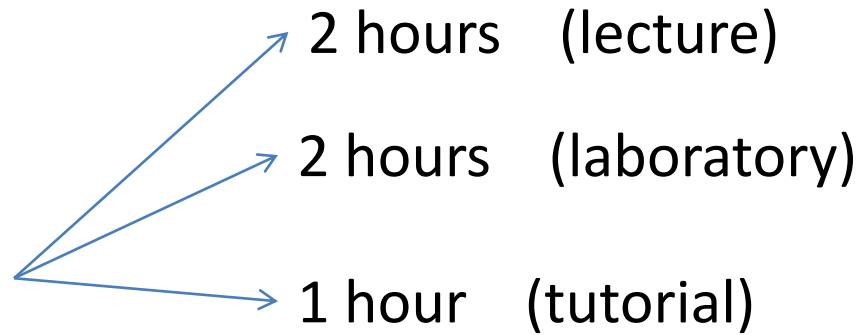
## CMP 410

### LECTURE 1

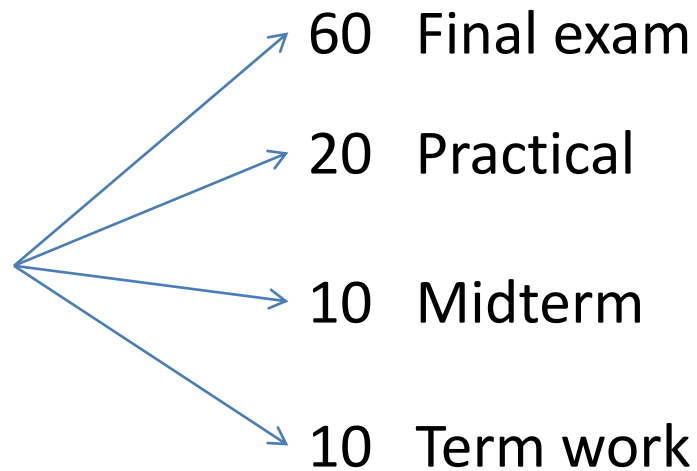
My site

<https://AssemSite8.wix.com/site8>

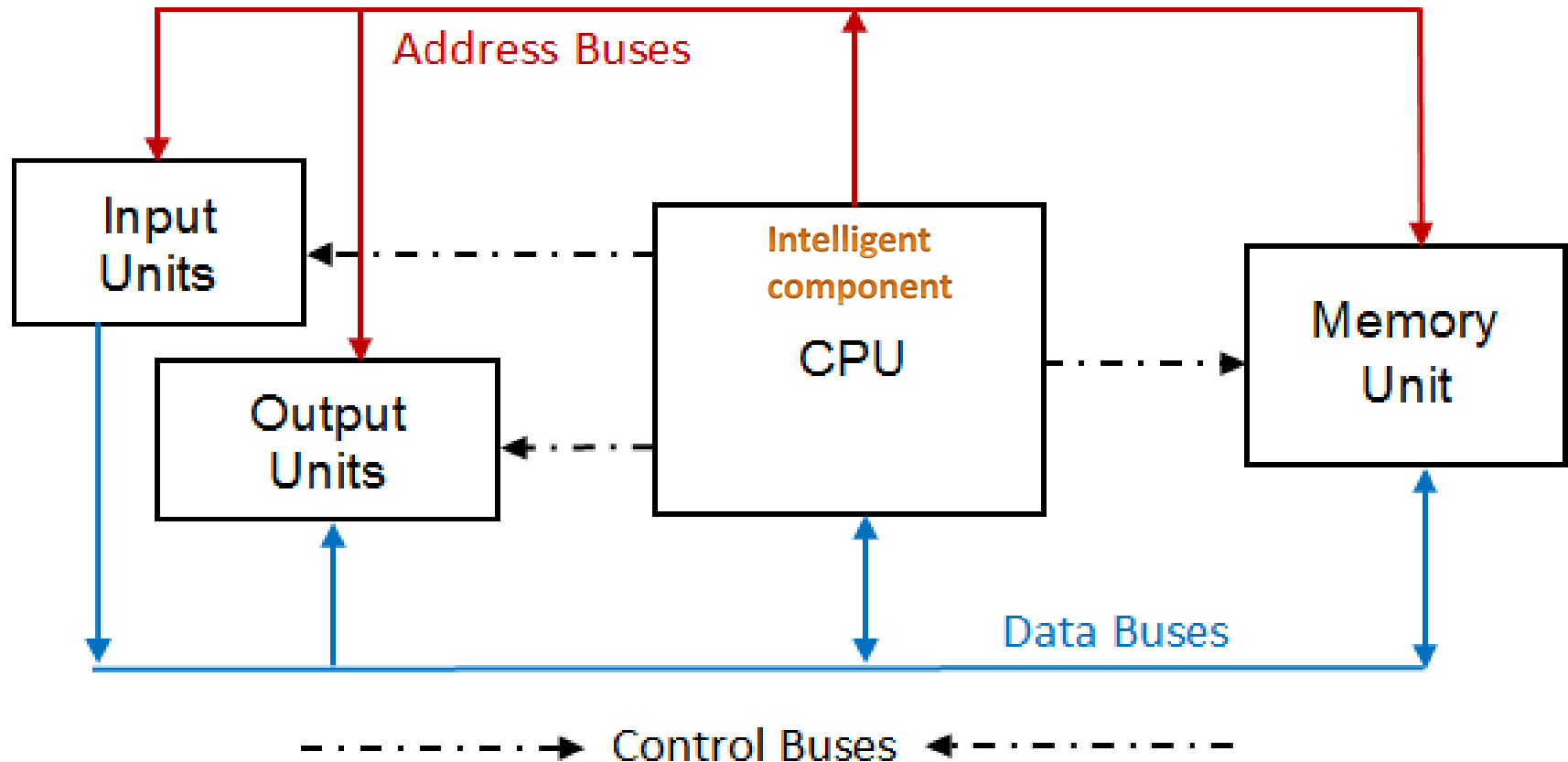
Every  
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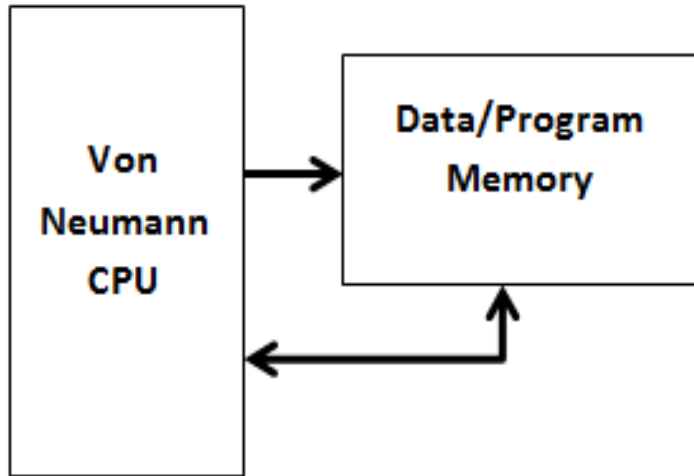
100  
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# The Basic Architecture of the Computer Systems

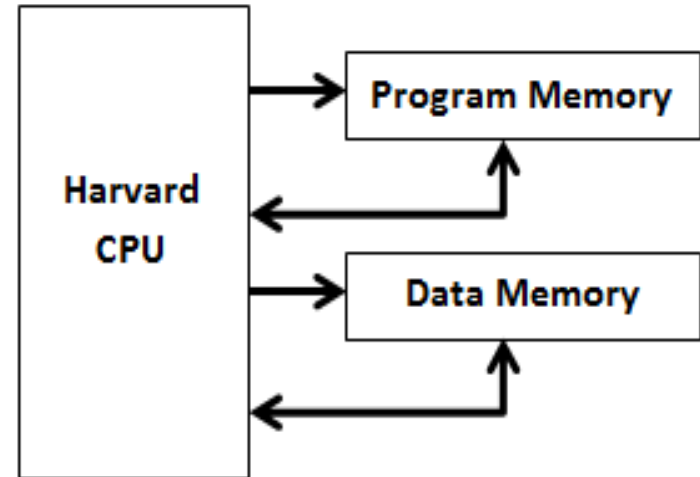


# The difference between Von Neumann and Harvard CPU architectures



It has single path for its memories

Likes microprocessor 8086



It has dual paths for its memories

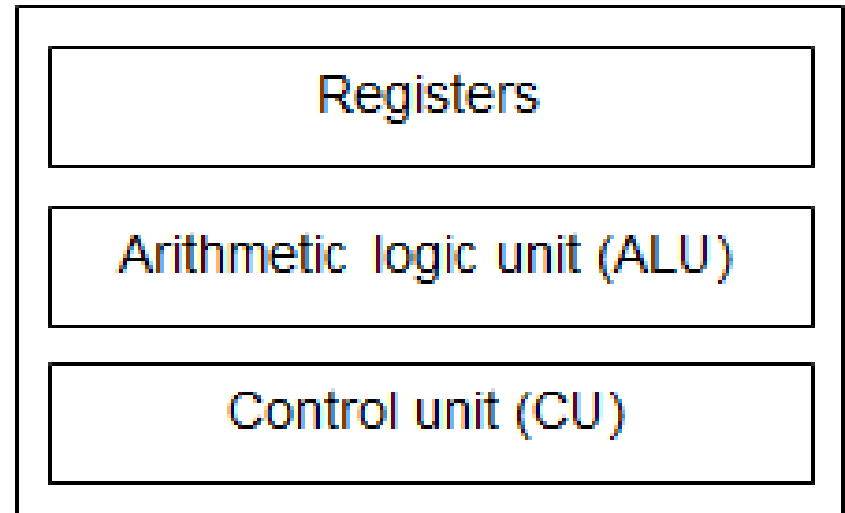
Likes the most microcontrollers such as 8051

# Microprocessor ( $\mu p$ )

it's a CPU in a single integrated microchip.

It has 3 major units:

- Arithmetic Logic Unit (ALU)
- Group of registers
- Control Unit (CU).



# Microcontroller ( $\mu$ C)

it's a CPU in a single integrated microchip

In addition the following:

- Data memory
- Program memory
- Data ports
- Timers
- ADC & DAC (recently)
- PWM (recently)

|       |       |                       |
|-------|-------|-----------------------|
| CPU   | RAM   | ROM                   |
| I / O | Timer | Serial<br>COM<br>Port |

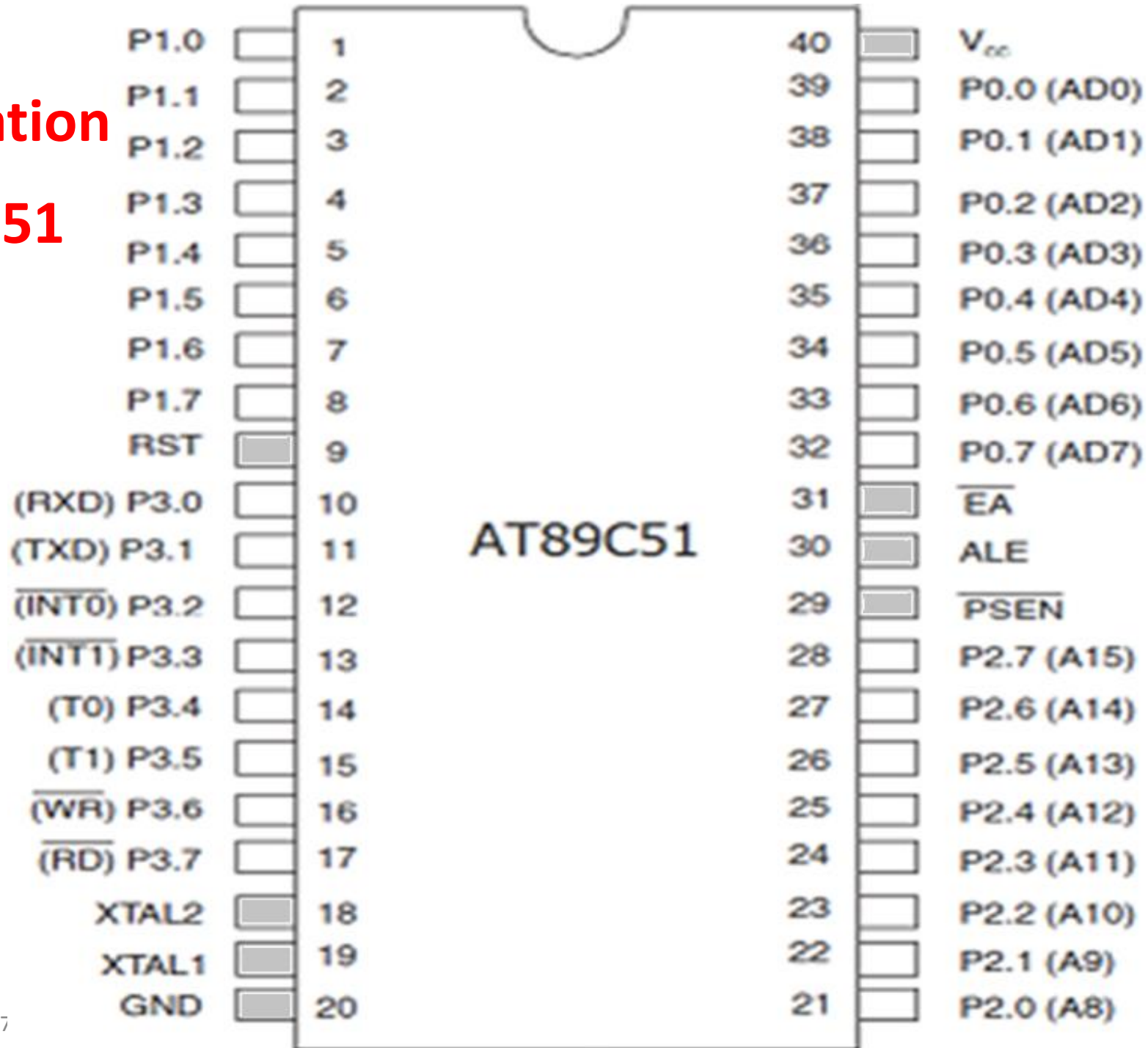
# Examples of the real Microcontrollers families

- ☐ PIC
- ☐ **MCS-51**
- ☐ ATmega
- ☐ ARM
- ☐ AVR

## AT89C51

- The AT89C51 is member of the MCS-51 Microcontroller family.
- It invented by Intel corporation
- Recently, it fabricated by Atmel and more than 20 corporations

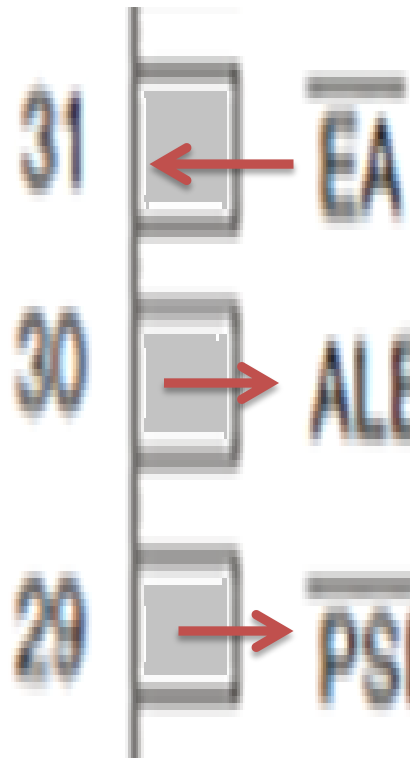
# The Pin configuration Of AT89C51



# The alternate functions of Port3

| Port 3 pins | Alternate-functions      | Descriptions                          |
|-------------|--------------------------|---------------------------------------|
| P3.0        | RXD                      | serial input pin                      |
| P3.1        | TXD                      | serial output pin                     |
| P3.2        | $\overline{\text{INT0}}$ | external interrupt 0 pin              |
| P3.3        | $\overline{\text{INT1}}$ | external interrupt 1 pin              |
| P3.4        | T0                       | timer 0 external input pin            |
| P3.5        | T1                       | timer 1 external input pin            |
| P3.6        | $\overline{\text{WR}}$   | external data memory write enable pin |
| P3.7        | $\overline{\text{RD}}$   | external data memory read enable pin  |

External access pin to select either the internal or external program memory

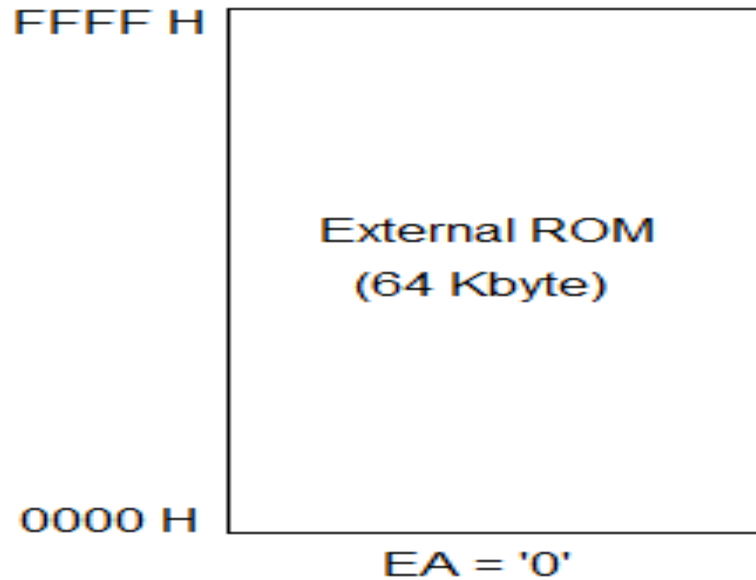


Address latch enable pin to separate the address bus from the address/data bus.

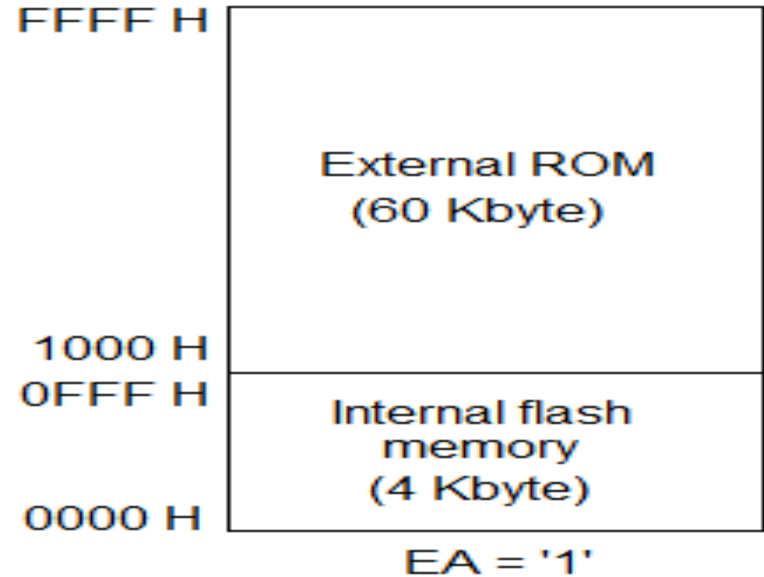
Program strobe enable pin to connect with the chip select (CS) pin of external ROM.

# The effecting of pin “EA” on the program memory

EA (External Access)

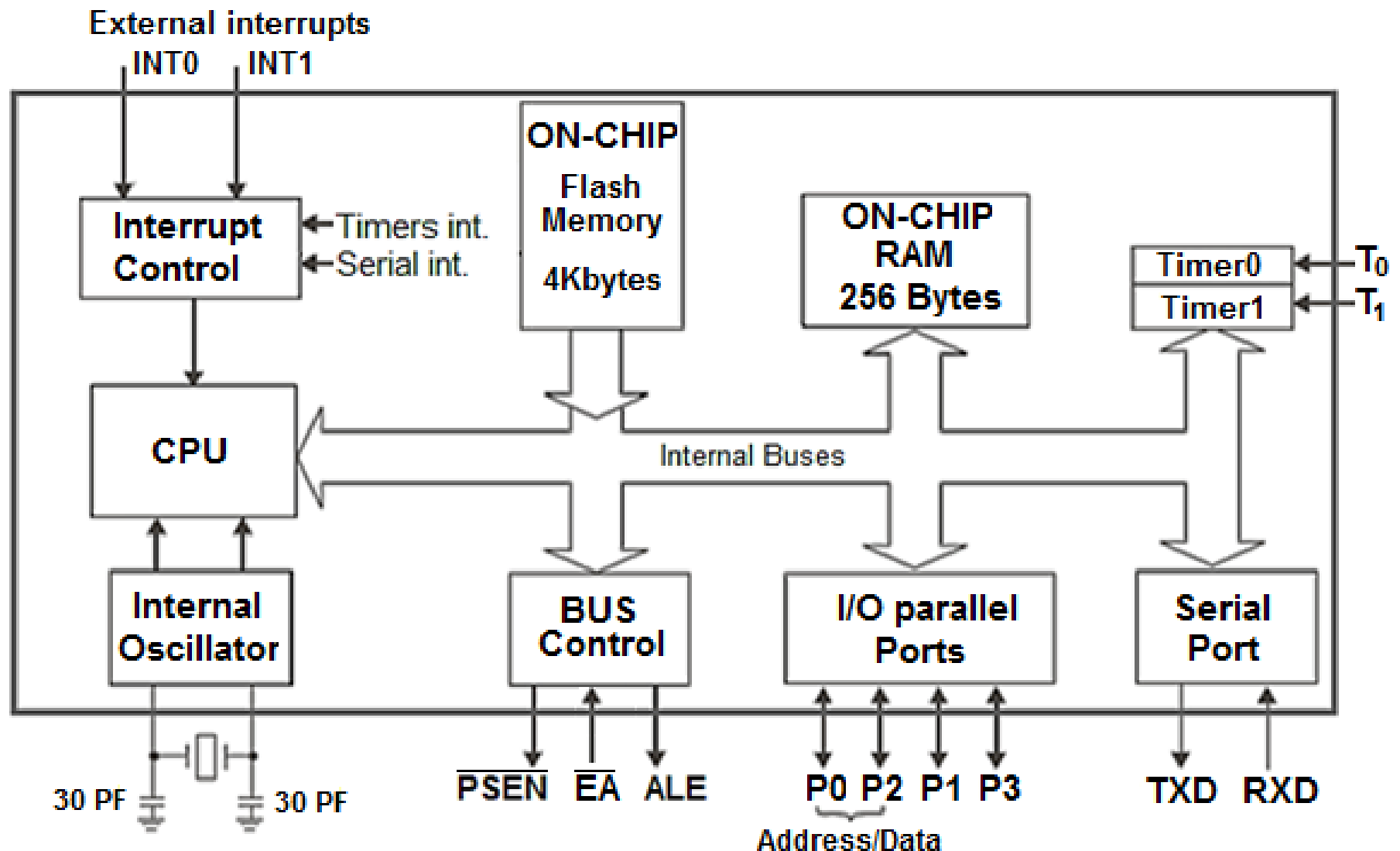


If the (EA = 0) then the AT89c51 will fetch their instructions from the external program memory only.



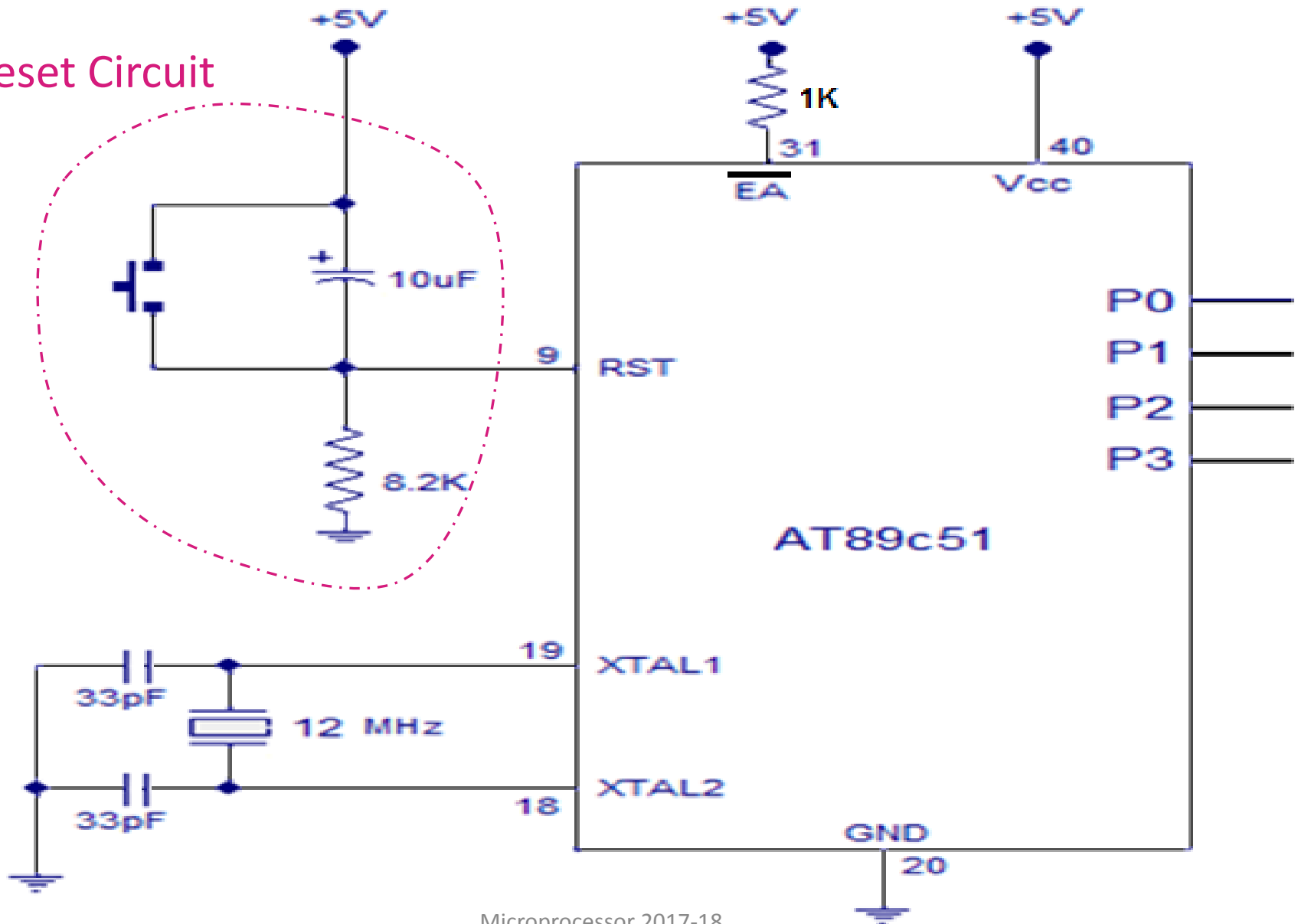
If the (EA = 1) then the AT89c51 will fetch their instructions from the internal program memory and follows by external one.

# The internal architecture of AT89C51



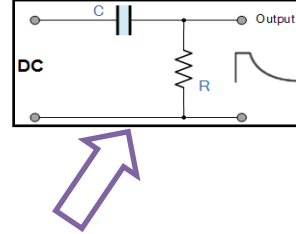
# The Basic Circuit for the AT89C51

Reset Circuit



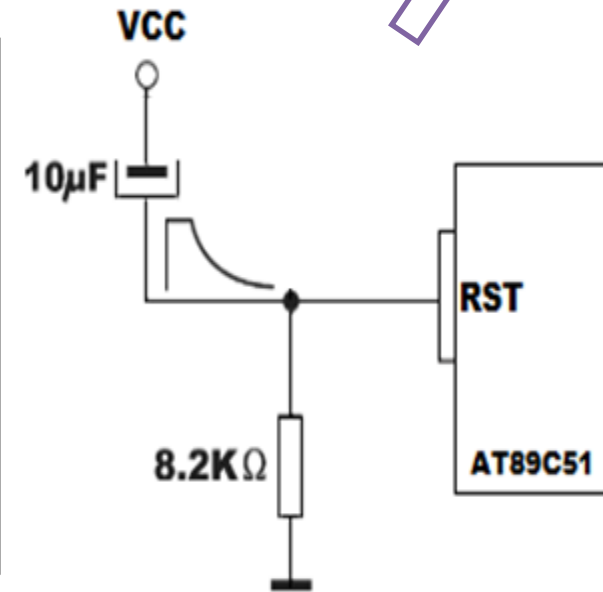
# Types of resetting

Differential circuit



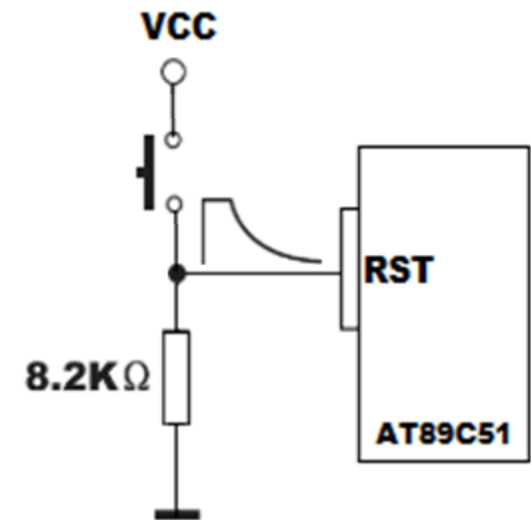
## 1- Automatic restarting

After Powering-on the 8051, positive pulse (with time duration  $> 2$  machine cycles) will apply to pin “pin 9”.



## 2- Manual restarting

After pressing the push-button, positive pulse will apply to reset-pin “pin 9”.



# After resetting the AT89c51

The initial states of the AT89c51 after resetting it.

❑ All locations of internal RAM will load by “00h” except:-

1) All ports (P0, P1, P2, P3) will load by “FFh”.

2) Stack pointer (SP) will load by “07h”.

❑ The 16-bit program counter (PC) will load by “0000h”.

## Important note

In case of powering-on any microcontroller **without** resetting it, the PC will load by **random address**, therefore the CPU will start fetching any **random instruction** from the program memory.